ASSIGNMENT 1 PYTHON BASICS

SET A

1. **Write a Python program to demonstrate the use of comments (single-line and multi-line).**

```
# This is a single-line comment
# Single-line comments start with the # symbol
# They are used to explain a line or part of the code

# Program to add two numbers

num1 = 10      # First number
num2 = 20      # Second number

# Adding the two numbers
sum = num1 + num2

print("Sum =", sum)  # Display the result


"""
This is a multi-line comment.
Multi-line comments are written using triple quotes (''' or """).
They are generally used to write longer explanations
or to describe the purpose of a program.

This program demonstrates:
1. Single-line comments
2. Multi-line comments
"""

print("Program executed successfully")
```

2. **Write a python program to declare variables of different standard data types (int, float, string) and display their values.**

```

# Integer variable
num = 10

# Float variable
price = 99.5
```

```
# String variable
name = "Python Programming"

# Displaying the values
print("Integer value:", num)
print("Float value:", price)
print("String value:", name)
```

3. Write a Python program to display “Good Morning!” on the screen.

```
print("Good Morning!")
```

4. Write a program to take the user’s name as input and print a greeting message to user.

```
name = input("Enter your name: ")
print("Hello", name, "! Welcome!")
```

5. Write a program to demonstrate different number data types in Python.

```
# Integer data type
a = 10
print("Integer value:", a, "Type:", type(a))

# Float data type
b = 25.75
print("Float value:", b, "Type:", type(b))

# Complex data type
c = 3 + 4j
print("Complex value:", c, "Type:", type(c))
```

SET B

1. **Write a Python program to show proper indentation using an if statement.**

```
num = 10

if num > 5:
    print("Number is greater than 5")
    print("This statement is inside the if block")

print("This statement is outside the if block")
```

2. **Write a program to display the type of each variable using the type() function.**

```
a = 10
b = 25.5
c = "Python"
d = 3 + 2j

print("Value of a:", a, "Type:", type(a))
print("Value of b:", b, "Type:", type(b))
print("Value of c:", c, "Type:", type(c))
print("Value of d:", d, "Type:", type(d))
```

3. **Write a program to take multiple inputs from the user in a single line**

```
a, b, c = input("Enter three values separated by space: ").split()

print("First value:", a)
print("Second value:", b)
print("Third value:", c)
```

4. **Write a Python program to accept user input for a list of numbers and display the list.**

Method 1.

```
numbers = input("Enter numbers separated by space: ").split()
```

```
# Convert each input value to integer
numbers = list(map(int, numbers))

print("The list of numbers is:", numbers)
```

Method 2.

```
numbers = []
n = int(input("How many numbers? "))

for i in range(n):
    num = int(input("Enter number: "))
    numbers.append(num)

print("The list of numbers is:", numbers)
```

5. Write a Python program that takes student details (name, roll number, marks) as input and displays them in a formatted manner.

Method 1.

```
name = input("Enter student name: ")
roll_no = input("Enter roll number: ")
marks = float(input("Enter marks: "))

print("\n--- Student Details ---")
print("Name       :", name)
print("Roll No.   :", roll_no)
print("Marks      :", marks)
```

Method 2.

```
name = input("Enter student name: ")
roll_no = input("Enter roll number: ")
marks = float(input("Enter marks: "))
print(f"\nName: {name}\nRoll No.: {roll_no}\nMarks: {marks}")
```

SET C

1. Write a Python program to display all keys and values of a dictionary

```
student = {
    "Name": "Amit",
    "Roll No": 23,
    "Marks": 88
}

for key in student:
    print(key, ":", student[key])
```

Method 2.

```
student = {
    "Name": "Amit",
    "Roll No": 23,
    "Marks": 88
}

print("Keys:")
for key in student.keys():
    print(key)

print("\nValues:")
for value in student.values():
    print(value)
```

2. Write a program to create a tuple of five elements and display it

```
my_tuple = (10, 20, 30, 40, 50)

print("Tuple elements are:", my_tuple)
```

ASSIGNMENT 2OPERATORS AND CONTROL STRUCTURES

SET A

1. **Write a Python program to find the sum and percentage of the marks obtained in 3 subjects.**

```
m1 = float(input("Enter marks of subject 1: "))
m2 = float(input("Enter marks of subject 2: "))
m3 = float(input("Enter marks of subject 3: "))

total = m1 + m2 + m3
percentage = (total / 300) * 100

print("Total Marks:", total)
print("Percentage:", percentage)
```

2. **Write a program to check whether a number is even or odd.**

```
num = int(input("Enter a number: "))

if num % 2 == 0:
    print("The number is Even")
else:
    print("The number is Odd")
```

3. **Write a program to find sum of a digit of a number.**

```
num = int(input("Enter a number: "))
sum = 0

while num > 0:
    digit = num % 10
    sum += digit
    num //= 10

print("Sum of digits:", sum)
```

4. **Write a Python program to check whether a given number is positive, negative, or zero.**

```
num = int(input("Enter a number: "))

if num > 0:
```

```
    print("Positive number")
elif num < 0:
    print("Negative number")
else:
    print("Zero")
```

5. Write a program to display numbers from 1 to 10 using a for loop.

```
for i in range(1, 11):
    print(i)
```

SET B

1. Write a Python program to find the largest of three numbers using nested if-else statements.

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
c = int(input("Enter third number: "))

if a > b:
    if a > c:
        print("Largest number is:", a)
    else:
        print("Largest number is:", c)
else:
    if b > c:
        print("Largest number is:", b)
    else:
        print("Largest number is:", c)
```

2. Write a Python program to check that two strings are equal or not. Print appropriate message.

```
str1 = input("Enter first string: ")
str2 = input("Enter second string: ")

if str1 == str2:
    print("Both strings are equal")
else:
    print("Strings are not equal")
```

3. Write a Python program to check whether a character is present in a given string or not.

```
string = input("Enter a string: ")
ch = input("Enter a character to search: ")

if len(ch) != 1:
    print("Please enter only ONE character")
else:
    if ch in string:
        print("Character is present in the string")
    else:
        print("Character is not present in the string")
```

4. Write a Python program to print all even numbers between 1 and 50 using a while loop.

```
i = 1

while i <= 50:
    if i % 2 == 0:
        print(i)
    i += 1
```

5. Write a Python program to check number is Perfect or not.
(Perfect number = sum of its proper divisors equals the number)

```
num = int(input("Enter a number: "))
sum = 0

for i in range(1, num):
    if num % i == 0:
        sum += i

if sum == num:
    print("Perfect number")
else:
    print("Not a perfect number")
```

6. Write a Python program to find factorial of a number.

```
num = int(input("Enter a number: "))
fact = 1

for i in range(1, num + 1):
    fact = fact * i

print("Factorial is:", fact)
```

SET C

1. Write a program to find the Fibonacci series.

First two numbers are 0 and 1
Each next number = sum of previous two

```
n = int(input("Enter how many terms: "))

a = 0
b = 1

print("Fibonacci series:")
print(a, b, end=" ")

for i in range(2, n):
    c = a + b
    print(c, end=" ")
    a = b
    b = c
```

2. Write a Python program to find number is Armstrong or not.

(Armstrong number = sum of cubes of digits equals the number)
Example: $153 = 1^3 + 5^3 + 3^3$

```
num = int(input("Enter a number: "))
temp = num
sum = 0

while temp > 0:
    digit = temp % 10
    sum = sum + digit ** 3
    temp = temp // 10

if sum == num:
    print("Armstrong number")
else:
    print("Not an Armstrong number")
```

ASSIGNMENT 3 PYTHON STRINGS

SET A

1. **Write a program to replace all occurrences of “a“ with $ in a String. (Ex. apple then output is $pple).**

```
s = input("Enter a string: ")
print(“Original string:”, s)
result = s.replace('a', '$')
print("Modified string:", result)
```

2. **Write a python program to Reverse words in a given String**

```
s = input("Enter a string: ")
words = s.split()

reversed_words = []
for word in words:
    reversed_words.append(word[::-1])

result = " ".join(reversed_words)
print("Reversed words string:", result)
```

3. **Write a Python program to count the number of characters (character frequency) in a string. Sample String: ‘google.com' Expected Result : {'o': 3, 'g': 2, '.': 1, 'e': 1, 'l': 1, 'm': 1, 'c': 1}**

```
s = input("Enter a string: ")
freq = {}

for ch in s:
    if ch in freq:
        freq[ch] += 1
    else:
        freq[ch] = 1

print("Character Frequency:", freq)
```

4. **Write a python program to check whether the string is Palindrome**

```
s = input("Enter a string: ")

if s == s[::-1]:
```

```python
    print("Palindrome")
else:
    print("Not Palindrome")
```

5. Write a Python program to calculate the Length of a String without using a Library Function.

```python
s = input("Enter a string: ")
count = 0

for ch in s:
    count += 1

print("Length of string:", count)
```

6. Calculate the sum and average of the digits present in a string

```python
s = input("Enter a string: ")
total = 0
count = 0

for ch in s:
    if ch.isdigit():
        total += int(ch)
        count += 1

if count != 0:
    avg = total / count
    print("Sum of digits:", total)
    print("Average of digits:", avg)
else:
    print("No digits found")
```

SET B

1. Write a python program to print even length words in a string

Accept a string
Split string into words
Check length of each word
Print word if length is even

```
s = input("Enter a string: ")
words = s.split()

print("Even length words are:")
for word in words:
    if len(word) % 2 == 0:
        print(word)
```

2. Write a python program to accept the strings which contains all vowels

Check whether a string contains **all five vowels**
(a, e, i, o, u)
Convert string to lowercase
Check presence of each vowel
If all present → valid string

```
s = input("Enter a string: ").lower()

vowels = {'a', 'e', 'i', 'o', 'u'}

if vowels.issubset(set(s)):
    print("String contains all vowels")
else:
    print("String does NOT contain all vowels")
```

3. Write a python program to Count the Number of matching characters in a pair of string

Count how many **common characters** appear in both strings
(No duplicates counted twice)
Convert both strings into sets
Find intersection
Count matching characters

```
s1 = input("Enter first string: ")
```

```
s2 = input("Enter second string: ")

matching = set(s1) & set(s2)

print("Matching characters:", matching)
print("Number of matching characters:", len(matching))
```

4. Write a python program to arrange string characters such that lowercase letters should come first

Separate lowercase characters
Separate uppercase characters
Combine them

```
s = input("Enter a string: ")

lower = ""
upper = ""

for ch in s:
    if ch.islower():
        lower += ch
    else:
        upper += ch

result = lower + upper
print("Rearranged string:", result)
```

5. Write a python program to replace special symbol in string with # character.

Traverse string
If character is not alphanumeric → replace with #

```
s = input("Enter a string: ")
result = ""

for ch in s:
    if ch.isalnum():
        result += ch
    else:
        result += "#"

print("Modified string:", result)
```

6. Write a python program to convert half of the string to uppercase
Input : python Output : PYThon or python

Find middle index
Convert either:First half to uppercase ORSecond half to uppercase

```
s = input("Enter a string: ")
mid = len(s) // 2

result = s[:mid].upper() + s[mid:]
print("Result:", result)
```

SET C

1. Write a python program to find word in string which contain both digit and number

```
s = input("Enter a string: ")

words = s.split()
result = []

for word in words:
    has_alpha = False
    has_digit = False

    for ch in word:
        if ch.isalpha():
            has_alpha = True
        elif ch.isdigit():
            has_digit = True

    if has_alpha and has_digit:
        result.append(word)

print("Words containing both letters and digits:")
print(result)
```

2. Write a Python program to compress a string as follows:
Example: Input: aaabbccccd
Output: a3b2c4d1

```
s = input("Enter a string: ")
compressed = ""
count = 1

for i in range(len(s) - 1):
    if s[i] == s[i + 1]:
        count += 1
    else:
        compressed += s[i] + str(count)
        count = 1

# for the last character
compressed += s[-1] + str(count)

print("Compressed string:", compressed)
```

ASSIGNMENT 4 PYTHON FUNCTIONS

SET A

1. **Write a Python function to find the Max of three numbers.**

```
def max_of_three(a, b, c):
    return max(a, b, c)

print(max_of_three(10, 25, 15))
```

2. **Write a Python program to reverse a string.**

```
def reversed_string(s):
  return s[::-1]

s = input("Enter a string: ")
print("Reversed string:", reversed_string(s))
```

3. **Write an anonymous function to calculate area of square.**

```
area_square = lambda side: side * side
print("Area of square:", area_square(5))
```

4. **Write a Python function to check whether a number is in a given range.**

```
def is_in_range(num, start, end):
    return start <= num <= end

print(is_in_range(7, 1, 10))
```

5. **Write a Python function that accepts a string and calculate the number of uppercase lettersand lower case letters.**

```
def count_case(s):
    upper = 0
    lower = 0

    for ch in s:
        if ch.isupper():
            upper += 1
        elif ch.islower():
            lower += 1
```

```
    print("Uppercase letters:", upper)
    print("Lowercase letters:", lower)

count_case("Hello World PYTHON")
```

6. Write a python functions to calculate the area of different geometric shapes (e.g., circle, rectangle,triangle) based on provided dimensions.

```
import math

def area_circle(radius):
    return math.pi * radius * radius

def area_rectangle(length, width):
    return length * width

def area_triangle(base, height):
    return 0.5 * base * height

print("Area of Circle:", area_circle(5))
print("Area of Rectangle:", area_rectangle(4, 6))
print("Area of Triangle:", area_triangle(3, 8))
```

7. Write python program that inputs a number and prints the multiplication table of that numberusing function.

```
def multiplication_table(n):
    for i in range(1, 11):
        print(n, "x", i, "=", n * i)

num = int(input("Enter a number: "))
multiplication_table(num)
```

SET B

1. **Write a python program that converts a decimal number to binary number usingfunction.**

```
def decimal_to_binary(n):
    return bin(n)[2:]   # removes '0b' prefix

num = int(input("Enter a decimal number: "))
print("Binary number:", decimal_to_binary(num))
```

2. **Write a python program to find the sum of natural number using recursive function.**

```
def sum_natural(n):
    if n == 0:
        return 0
    else:
        return n + sum_natural(n - 1)

num = int(input("Enter a number: "))
print("Sum of first", num, "natural numbers:", sum_natural(num))
```

3. **Write a python program to print the Fibonacci sequence using recursive function**

```
def fibonacci(n):
    if n == 0:
        return 0
    elif n == 1:
        return 1
    else:
        return fibonacci(n - 1) + fibonacci(n - 2)

terms = int(input("Enter number of terms: "))

print("Fibonacci sequence:")
for i in range(terms):
    print(fibonacci(i), end=" ")
```

4. **Write a Python function to check whether a number is perfect or not.**

```
def is_perfect(n):
    if n <= 1:
```

```
        return False

    sum_div = 0
    for i in range(1, n):
        if n % i == 0:
            sum_div += i

    return sum_div == n

num = int(input("Enter a number: "))
if is_perfect(num):
    print(num, "is a Perfect Number")
else:
    print(num, "is not a Perfect Number")
```

5. Write a Python function that takes a number as a parameter and check the numberis prime or not.

```
def is_prime(n):
    if n <= 1:
        return False
    for i in range(2, int(n**0.5) + 1):
        if n % i == 0:
            return False
    return True

num = int(input("Enter a number: "))
if is_prime(num):
    print(num, "is a Prime Number")
else:
    print(num, "is not a Prime Number")
```

SET C

1. **Create an inner function to calculate the addition in the following way**
    a. **Create an outer function that will accept two parameters a and b**
    b. **Create an inner function inside an outer function that will calculate the addition of a and b.**
    c. **At last, an outer function will add 5 into addition and return it.**

```
def outer_function(a, b):
    # Inner function
    def inner_function():
        return a + b

    # Call inner function and add 5 to its result
    result = inner_function() + 5
    return result


# Example
print(outer_function(10, 20))   # (10 + 20) + 5 = 35
```

2. **Write a Python program using separate functions to perform the following operations: Addition , Subtraction, Multiplication, Division, Modulus Use a menu-driven approach and call appropriate functions based on user choice.**

```
# Separate functions
def add(a, b):
    return a + b

def subtract(a, b):
    return a - b

def multiply(a, b):
    return a * b

def divide(a, b):
    if b == 0:
        return "Division by zero is not allowed"
    return a / b
```

```
def modulus(a, b):
    if b == 0:
        return "Modulus by zero is not allowed"
    return a % b


# Menu-driven program
while True:
    print("\n----- MENU -----")
    print("1. Addition")
    print("2. Subtraction")
    print("3. Multiplication")
    print("4. Division")
    print("5. Modulus")
    print("6. Exit")

    choice = int(input("Enter your choice (1-6): "))

    if choice == 6:
        print("Exiting program...")
        break

    a = float(input("Enter first number: "))
    b = float(input("Enter second number: "))

    if choice == 1:
        print("Result =", add(a, b))
    elif choice == 2:
        print("Result =", subtract(a, b))
    elif choice == 3:
        print("Result =", multiply(a, b))
    elif choice == 4:
        print("Result =", divide(a, b))
    elif choice == 5:
        print("Result =", modulus(a, b))
    else:
        print("Invalid choice! Please select from 1 to 6.")
```

ASSIGNMENT 5 PYTHON LISTS

SET A

1. **Write a Python program to sum all the items in a list.**

```
lst = [10, 20, 30, 40]
total = 0
for i in lst:
    total += i
print("Sum =", total)
```

2. **Write a Python program to multiplies all the items in a list.**

```
lst = [1, 2, 3, 4]
product = 1
for i in lst:
    product *= i
print("Product =", product)
```

3. **Write a Python program to get the largest number from a list.**

```
lst = [45, 12, 78, 34, 90]
largest = lst[0]
for i in lst:
    if i > largest:
        largest = i
print("Largest =", largest)
```

4. **Write a Python program to get the smallest number from a list.**

```
lst = [45, 12, 78, 34, 90]
smallest = lst[0]
for i in lst:
    if i < smallest:
        smallest = i
print("Smallest =", smallest)
```

5. **Write a Python program to check if a list is empty or not**

```
lst = []

if len(lst) == 0:
    print("List is empty")
else:
    print("List is not empty")
```

SET B

1. Write a Python program to remove duplicates from a list.

```
lst = [1, 2, 3, 2, 4, 1, 5]
new_list = []
for i in lst:
    if i not in new_list:
        new_list.append(i)
print("After removing duplicates:", new_list)
```

2. Write a Python program to copy a list.

```
lst = [10, 20, 30, 40]
copy_list = lst.copy()
print("Original list:", lst)
print("Copied list:", copy_list)
```

3. Write a Python program that removes all even numbers from a specified list and displays the remaining elements.

```
lst = [1, 2, 3, 4, 5, 6, 7, 8]
result = []
for i in lst:
    if i % 2 != 0:
        result.append(i)
print("After removing even numbers:", result)
```

4. Write a Python program to append a list to the second list.

```
list1 = [1, 2, 3]
list2 = [4, 5, 6]
list2.extend(list1)
print("After appending:", list2)
```

5. Write a Python Program to Square Each Element of the List and Print List in Reverse Order

```
lst = [1, 2, 3, 4, 5]
square_list = []
for i in lst:
    square_list.append(i * i)
```

```
square_list.reverse()
print("Squared and reversed list:", square_list)
```

6. Write a Python Program to Remove All Occurrences of an Element from a Given List

```
lst = [1, 2, 3, 2, 4, 2, 5]
element = 2

result = []
for i in lst:
    if i != element:
        result.append(i)

print("List after removing all occurrences of", element, ":", result)
```

SET C

1. Write a Python Program to Interchange First and Last Elements of in a List

```
lst = [10, 20, 30, 40, 50]

lst[0], lst[-1] = lst[-1], lst[0]

print("List after interchange:", lst)
```

2. Write a Python Program to Remove Multiple Empty Strings from a List of Strings

```
lst = ["Python", "", "Java", "", "", "C", ""]

result = []
for i in lst:
    if i != "":
        result.append(i)

print("List after removing empty strings:", result)
```

ASSIGNMENT 6 PYTHON TUPLE

SET A

1. **Reverse the following tuple Tup=(15,25,35,45,55)**

```
Tup = (15, 25, 35, 45, 55)
rev = Tup[::-1]
print("Reversed tuple:", rev)
```

2. **Write a Python program to create a list of tuples with the first element as the number and second element as the square of the number.**

```
lst = []
for i in range(1, 6):
    lst.append((i, i*i))
print("List of tuples:", lst)
```

3. **Write a Python program to create a tuple with numbers and print one item.**

```
t = (10, 20, 30, 40, 50)
print("One item from tuple:", t[2])
```

4. **Write a Python program to unpack a tuple in several variables.**

```
t = (100, 200, 300)
a, b, c = t
print("a =", a)
print("b =", b)
print("c =", c)
```

5. **Write a Python program to add an item in a tuple.**

(Tuples are immutable, so we convert to list and back)

```
t = (10, 20, 30)
lst = list(t)
lst.append(40)
t = tuple(lst)
print("Updated tuple:", t)
```

6. **Copy element 22 and 66 from the following tuple in to anewtuple tuple1 = (11, 22, 33, 44, 55, 66)**

```
tuple1 = (11, 22, 33, 44, 55, 66)
new_tuple = (tuple1[1], tuple1[5])
print("New tuple:", new_tuple)
```

SET B

1. **Write a Python program to convert a tuple to a string.**

```
t = ('P', 'y', 't', 'h', 'o', 'n')
string = ''.join(t)
print("String:", string)
```

2. **Sort the tuple: Tuple=(2, 4,6, 1, 4, 7.8, 2.7)**

```
Tup = (2, 4, 6, 1, 4, 7.8, 2.7)
sorted_tup = tuple(sorted(Tup))
print("Sorted tuple:", sorted_tup)
```

3. **Write a Python program to get the 4th element from front and 6th element from last of a tuple.**

```
t = (10, 20, 30, 40, 50, 60, 70, 80)

print("4th element from front:", t[3])
print("6th element from last:", t[-6])
```

SET C

1. **Write a Python program to find the repeated items of a tuple.**

```
t = (1, 2, 3, 4, 2, 5, 3, 6, 1)
repeated = []

for i in t:
    if t.count(i) > 1 and i not in repeated:
        repeated.append(i)

print("Repeated items:", repeated)
```

2. **Write a Python program to check whether an element exists within a tuple**

```
t = (10, 20, 30, 40, 50)
element = 30

if element in t:
    print(element, "exists in the tuple")
else:
    print(element, "does not exist in the tuple")
```

ASSIGNMENT 7 PYTHON DICTIONARY

SET A

1. **Write a Python script to access the value of a key from a dictionary.**

```
my_dict = {'a': 10, 'b': 20, 'c': 30}

key = 'b'
print("Value of key", key, "is:", my_dict[key])
```

2. **Write a Python script to concatenate following dictionaries to create a new one.**
   Sample Dictionary:
   dic1={1:11,2:22}
   dic2={3:33,4:44}
   dic3={5:55,6:66}
   Expected Result: {1: 11,2: 22, 3:33, 4: 44, 5: 55,6: 66}

```

dic1 = {1:11, 2:22}
dic2 = {3:33, 4:44}
dic3 = {5:55, 6:66}

result = {}
result.update(dic1)
result.update(dic2)
result.update(dic3)

print(result)
```

3. **Write a Python program to iterate over dictionaries using for loops.**

```
my_dict = {'a':1, 'b':2, 'c':3}

for key in my_dict:
    print(key, ":", my_dict[key])
```

4. **Write a Python program to sum all the items in a dictionary.**
   Sample Dictionary : my_dict={'data1':100,'data2':-54,'data3':247}
   Expected Result: 293

```
my_dict = {'data1':100, 'data2':-54, 'data3':247}
```

```
total = sum(my_dict.values())
print("Sum:", total)
```

5. **Write a Python program to remove a key from a dictionary.**
   Sample Dictionary:my Dict={'a':1,'b':2,'c':3,'d':4}
   Sample Output:
   {'c':3,'b':2,'d':4, 'a':1}
   {'c':3,'b':2,'d':4}

```
myDict = {'a':1, 'b':2, 'c':3, 'd':4}

print(myDict)

# remove key 'a'
myDict.pop('a')

print(myDict)
```

SET B

1. **Write a Python program to sort a dictionary by key. Sample Dictionary:**
   color_dict={'red':'#FF0000','green':'#008000','black':'#000000','white':'#FFFFFF'}
   Expected Output:
   black:#000000green:#008000 red: #FF0000 white:#FFFFFF

```
color_dict =
{'red':'#FF0000','green':'#008000','black':'#000000','white':'#FFFFFF'}

for key in sorted(color_dict):
    print(key, ":", color_dict[key], end=" ")
```

2. **Write a Python program to combine two dictionary adding values for common keys.**
   Sample Dictionary:
   d1={'a':100,'b':200,'c':300}
   d2={'a':300,'b':200,'d':400}
   Sample out put: Counter({'a':400, 'b':400,'d':400,'c':300})

```

from collections import Counter

d1 = {'a':100,'b':200,'c':300}
d2 = {'a':300,'b':200,'d':400}

result = Counter(d1) + Counter(d2)
print(result)
```

3. **Write a Python script to generate and print a dictionary that contains a number (Between 1 and n) in the form (x, x*x).**
   Sample Dictionary (n=5)
   Expected Output : {1:1,2: 4, 3: 9, 4:16, 5: 25}

```
n = 5
my_dict = {}

for x in range(1, n+1):
    my_dict[x] = x * x

print(my_dict)
```

4. **Write a Python program to create a dictionary from a string.**
   Sample String:"W3resource"

Expected output:{'3':1,'s':1,'r':2, 'u':1,'w':1, 'c':1,'e':2,'o':1}

```
string = "W3resource"
my_dict = {}

for ch in string.lower():
    if ch in my_dict:
        my_dict[ch] += 1
    else:
        my_dict[ch] = 1

print(my_dict)
```

SET C

1. **Write a Python program to create a dictionary from two lists without losing duplicate values.**
   Samplelists:['Class-V','Class-VI','Class-VII','Class-VIII'],[1,2,2,3]
   Expected Output: default dict(<class 'set'>, {'Class-VII': {2}, 'Class-VI': {2}, 'Class-VIII':{3},'Class-V':{1}})

```
from collections import defaultdict

classes = ['Class-V','Class-VI','Class-VII','Class-VIII']
numbers = [1,2,2,3]

result = defaultdict(set)

for k, v in zip(classes, numbers):
    result[k].add(v)

print(result)
```

2. **Write a Python program to create a dictionary of keys x, y, and z where each key has as value a list from 11-20, 21-30, and 31-40 respectively. Access the fifth value of each key from the dictionary.**
   Expected Output:
   {'x':[11,12, 13,14,15, 16,17,18, 19],
   'y':[21,22, 23,24, 25,26, 27,28, 29],
   'z':[31, 32,33, 34,35, 36,37, 38, 39]}
   15
   25
   35
   xhas value[11, 12,13,14,15, 16,17, 18, 19]
   yhas value [21, 22, 23,24, 25, 26, 27, 28,29]
   z has value[31, 32,33, 34,35, 36,37, 38, 39]

```
my_dict = {
    'x': list(range(11, 20)),
    'y': list(range(21, 30)),
    'z': list(range(31, 40))
}

print(my_dict)

# Access 5th value (index 4)
print(my_dict['x'][4])
```

```
print(my_dict['y'][4])
print(my_dict['z'][4])

# Print full values
print("x has value", my_dict['x'])
print("y has value", my_dict['y'])
print("z has value", my_dict['z'])
```

ASSIGNMENT 8 PYTHON SET

SET A

1. **What is the output of following program: sets = {1, 2, 3, 4, 4} print(sets)**

```
sets = {1, 2, 3, 4, 4}
print(sets)
```

2. **Write a python program to remove and return an arbitrary set element. Raise Key Error if the set is empty**

```
my_set = {10, 20, 30, 40}

if len(my_set) == 0:
    raise KeyError("Set is empty")
else:
    removed = my_set.pop()
    print("Removed element:", removed)
    print("Remaining set:", my_set)
```

3. **Write a Python program to do iteration over sets.**

```
my_set = {1, 2, 3, 4, 5}

for item in my_set:
    print(item)
```

4. **Write a Python program to add and remove operation on set.**

```
my_set = {1, 2, 3}

# Add element
my_set.add(4)
print("After adding:", my_set)

# Remove element
my_set.remove(2)
print("After removing:", my_set)
```

SET B

1. **Write a Python program to accept the strings which contains all vowels.**

```
string = input("Enter a string: ").lower()

vowels = set('aeiou')

if vowels.issubset(set(string)):
    print("The string contains all vowels.")
else:
    print("The string does not contain all vowels.")
```

2. **Write a Python program to create a union of sets.**

```
set1 = {1, 2, 3}
set2 = {3, 4, 5}

union_set = set1.union(set2)
print("Union:", union_set)
```

3. **Write a Python program to create an intersection of sets.**

```

set1 = {1, 2, 3, 4}
set2 = {3, 4, 5, 6}

intersection_set = set1.intersection(set2)
print("Intersection:", intersection_set)
```

4. **Write a Python program to find maximum and the minimum value in a set.**

```
my_set = {10, 5, 30, 25, 15}

print("Maximum:", max(my_set))
print("Minimum:", min(my_set))
```

5. **Write a Python program to create set difference and asymmetric difference**

```
set1 = {1, 2, 3, 4}
set2 = {3, 4, 5, 6}

# Difference
```

```
diff = set1 - set2
print("Difference (set1 - set2):", diff)

# Symmetric difference
sym_diff = set1 ^ set2
print("Symmetric Difference:", sym_diff)
```

6. Write a Python program to find the length of a set.

```
my_set = {1, 2, 3, 4, 5}
print("Length of set:", len(my_set))
```

SET C

1. Write a Python program to perform different set operations.

```
A = {1, 2, 3, 4}
B = {3, 4, 5, 6}

print("Union:", A | B)
print("Intersection:", A & B)
print("Difference (A - B):", A - B)
print("Difference (B - A):", B - A)
print("Symmetric Difference:", A ^ B)
```

2. Write a Python program to create shallow copy of sets.

```
original_set = {1, 2, 3, 4}

# Shallow copy
copied_set = original_set.copy()

print("Original set:", original_set)
print("Copied set:", copied_set)
```